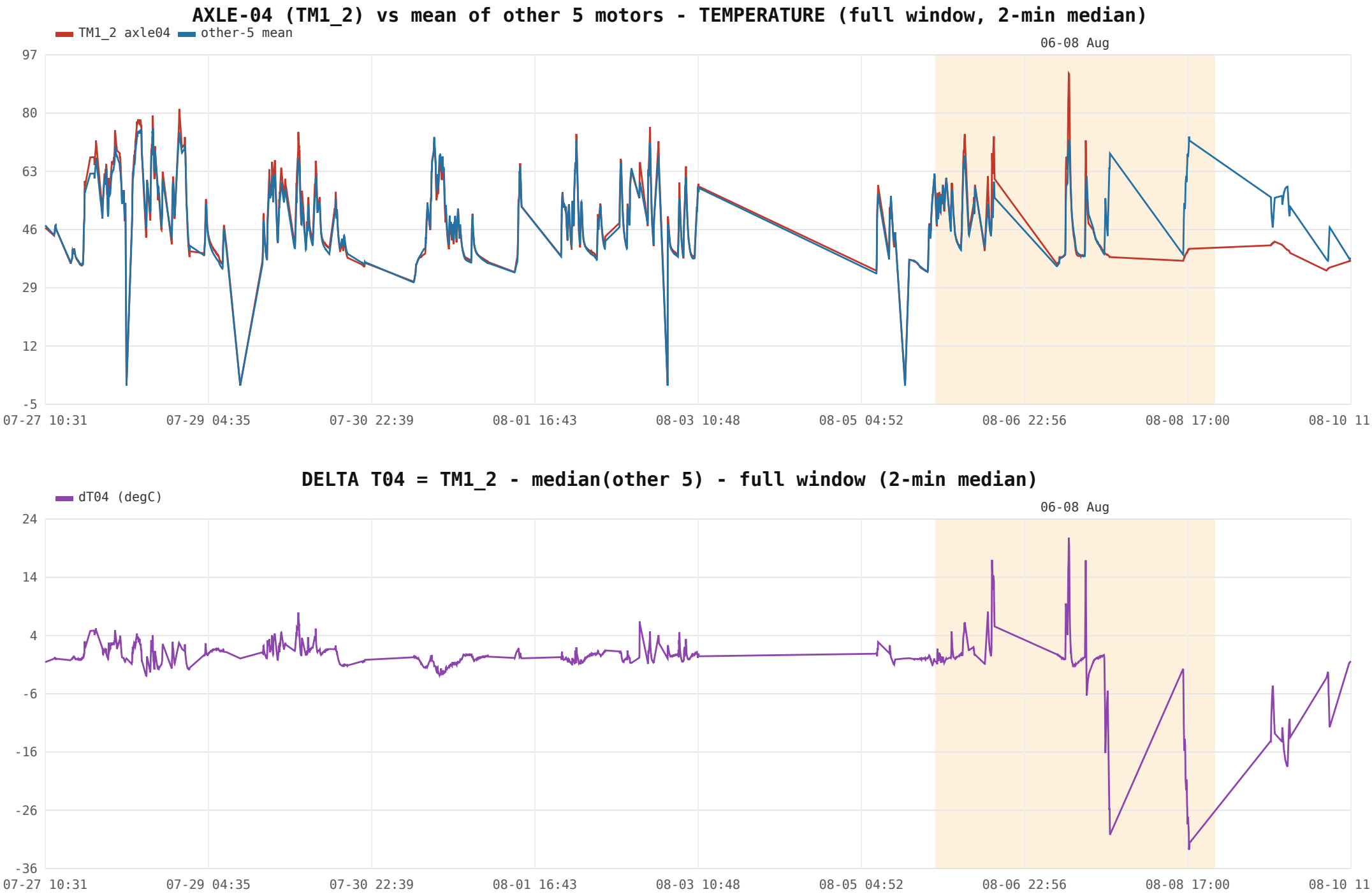


Loco 42728 – coupled-signal analysis (friction → resistance → traction effort → temperature)

Failure ground truth: 08-Aug-2026 19:26, axle locked – 2nd bogie, 04 axle (SinglePageFailureReport). Telemetry: 155,095 rows (Equus feed) 27-Jul→10-Aug 2026. Charts are SVG (matplotlib unavailable in this env).

Plot A – all six TM temperatures / Plot B – relative dT04



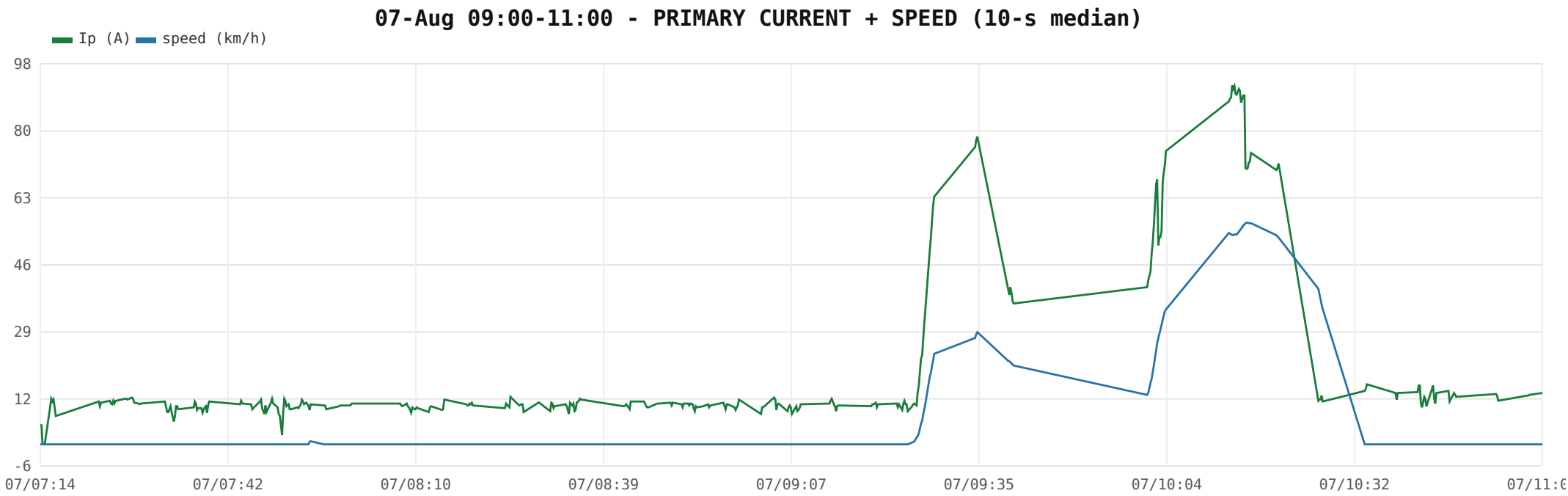
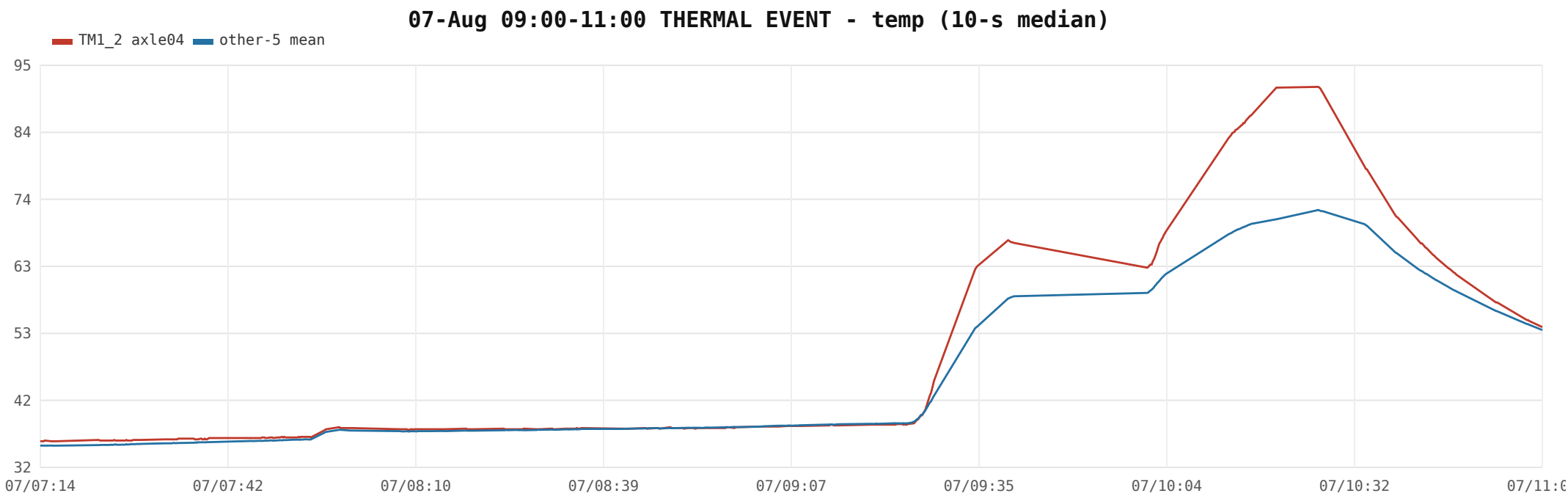
Read: through 05-Aug TM1_2 tracks the pack (dT04 ≈ 0). 06-Aug afternoon dT04 rises (+9...+10 °C). 07-Aug 10:13–10:34 massive spike to 91.7 °C (dT04 +19.7). On 08-Aug TM1_2 goes *cold* (≈ 37–40 °C, dT04 −27.7) while the other 5 motors run 60–74 °C – axle-04 no longer contributing, the remaining five carrying the train. Telemetry gap 08-Aug 17:25 → 09-Aug 14:25 covers the 19:26 lock event.

Electrical effort (primary current) + speed, full window



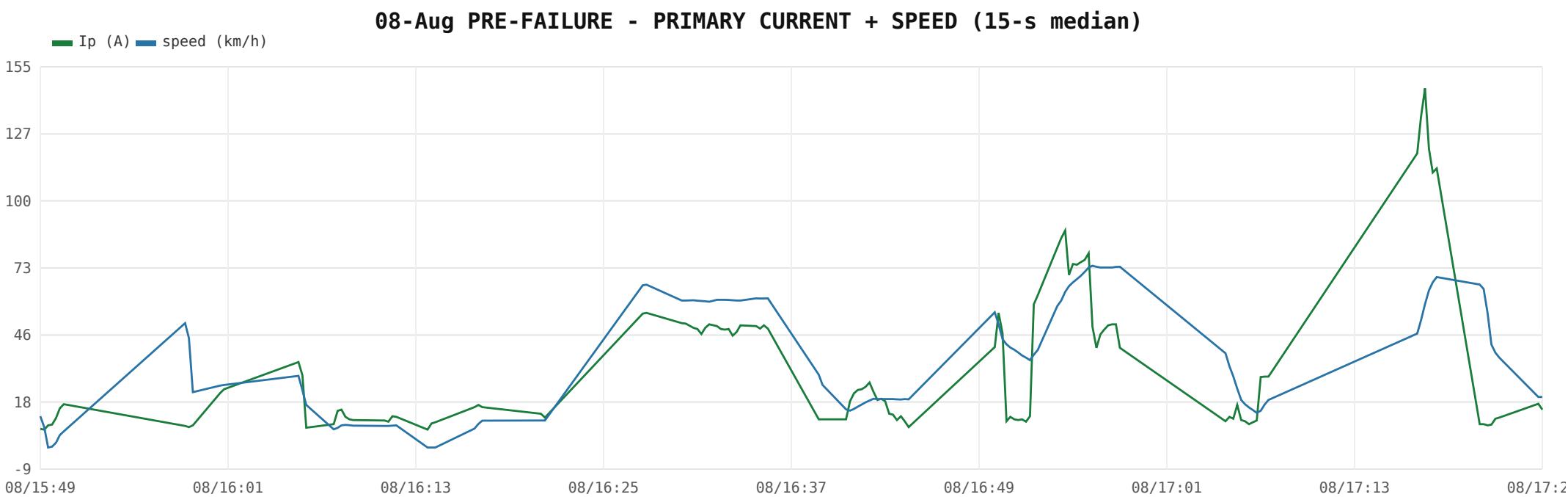
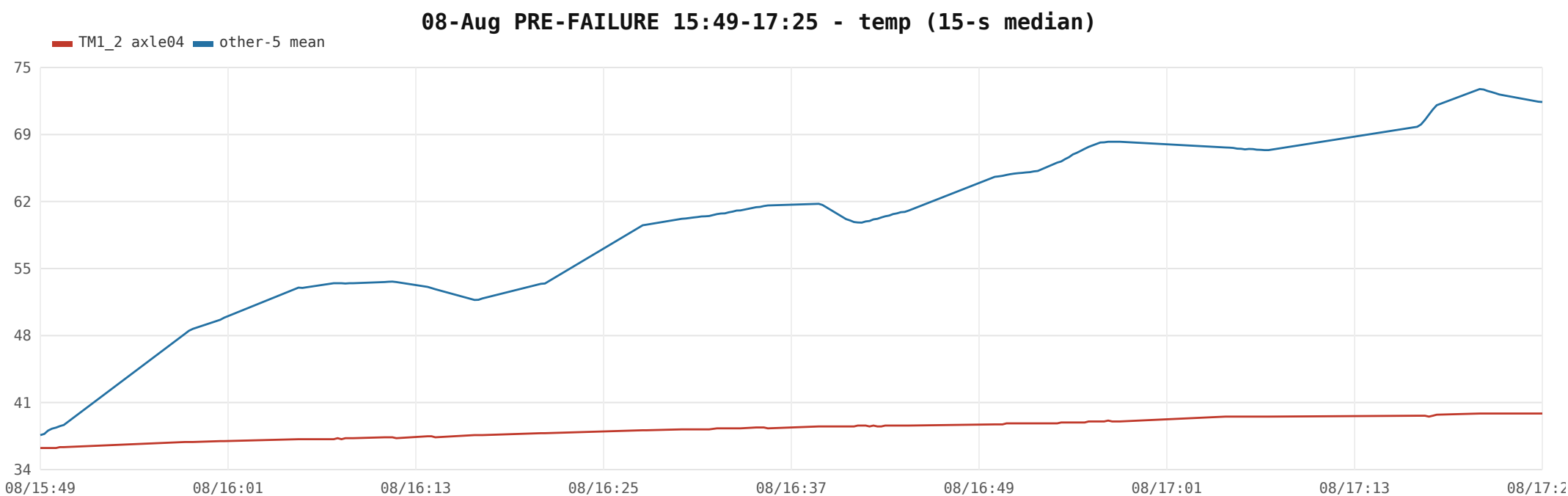
Read: primary current is *not* simply higher over time – it depends on speed/traction command/grade. The correct comparison is current *conditioned on speed AND acceleration* (Test B below). The energy counter (xenergymwh_ec) is a cumulative meter with unverified units – only the statement "the counter changes more rapidly during the failure-day running window" is safe; the raw kWh/min figure should NOT be used.

Window 3 – 07-Aug 10:00–10:40 thermal event (aligned signals)



Read: at the thermal peak (10:13–10:27, TM1_2 83–91.7 °C) the loco was doing ~53–55 km/h with Ip ≈ 70–92 A, then **speed decayed to ~40 km/h and to 0 by 10:34** as temperature crested – the locomotive could not hold speed while axle-04 overheated. All six bg*tm* IP voltages are equal (≈1713 V) – they are the shared DC-link, not a per-axle effort signal.

Window 4 – 08-Aug 15:49–17:25 pre-failure state



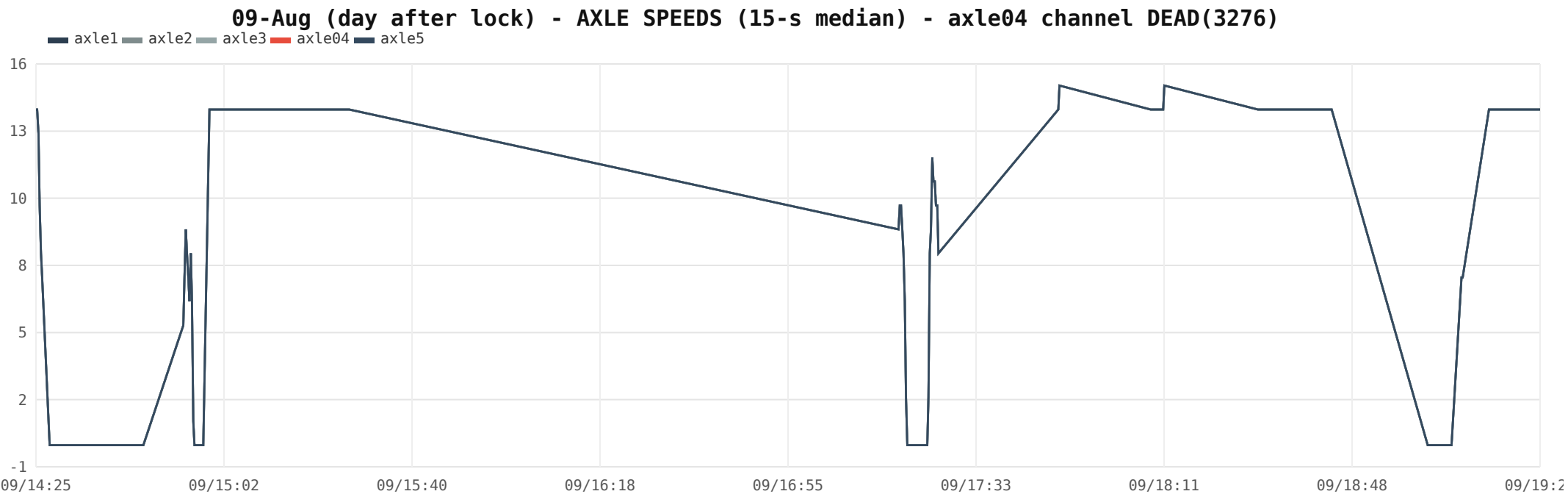
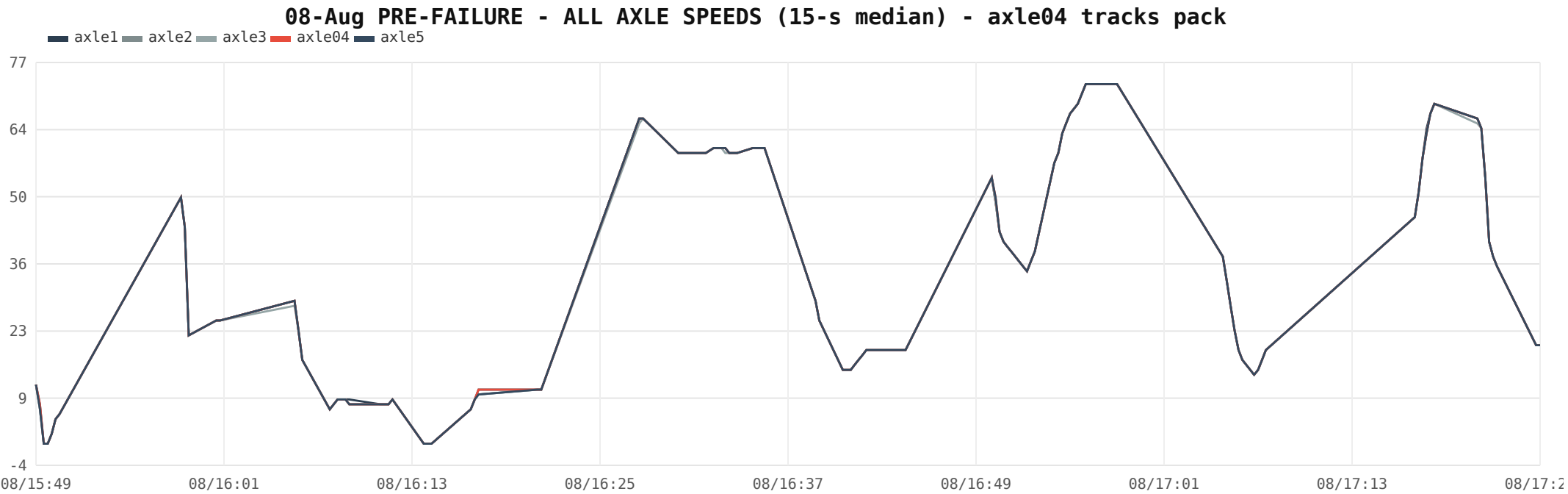
Read: axle-04 (TM1_2) is flat-cold $\approx 37\text{--}40\text{ }^{\circ}\text{C}$ for the whole session while the other five run $48\text{--}74\text{ }^{\circ}\text{C}$ and the loco accelerates to $60\text{--}73\text{ km/h}$ with primary current up to $130\text{--}144\text{ A}$ (17:17–17:18, accel to 2180 V DC-link). This is the state the 5-other-motors hypothesis predicts just before a lock: axle-04 motor cut out / unloaded, remaining traction system doing all the work.

TEST 1 – median primary current vs speed bucket (traction-active only)

period	bucket km/h	n	Ip med	Ip p90	TM1_2	other5	dT
06-Aug	0-10	631	13.2	22.0	50.6	49.1	+1.5
07-Aug	0-10	266	14.5	27.8	39.9	40.2	-0.3
08-Aug	0-10	169	12.7	16.6	37.6	53.4	-15.8
BASELINE 27Jul-03Aug	0-10	7603	12.5	20.8	45.9	45.9	-0.0
06-Aug	10-20	307	27.3	42.7	48.0	48.7	-0.7
07-Aug	10-20	107	41.0	54.4	43.5	43.1	+0.4
08-Aug	10-20	280	19.0	27.1	38.9	60.0	-21.1
BASELINE	10-20	5603	21.0	44.9	50.3	50.7	-0.4
06-Aug	20-30	274	29.5	67.9	53.0	50.6	+2.4
07-Aug	20-30	273	43.2	67.9	44.2	53.8	-9.7
08-Aug	20-30	35	18.6	23.7	40.0	71.8	-31.9
BASELINE	20-30	3513	37.4	81.3	56.8	56.0	+0.8
06-Aug	30-40	266	26.9	63.5	69.1	54.9	+14.2
07-Aug	30-40	69	65.7	74.5	67.5	61.3	+6.2
08-Aug	30-40	23	56.2	61.3	39.1	64.8	-25.7
BASELINE	30-40	2415	58.8	127.7	57.0	56.1	+0.9
06-Aug	40-50	32	88.4	93.0	60.4	60.5	-0.1
08-Aug	40-50	13	120.6	126.7	39.9	69.3	-29.5
BASELINE	40-50	2435	60.1	154.8	59.7	57.5	+2.2

Read: at comparable speed (30–40 km/h), 07-Aug median Ip (65.7) exceeds baseline (58.8) while axle-04 is the hot motor – the anomaly is coupled to higher primary current. On 08-Aug at 40–50 km/h the small sample shows median Ip 120.6 vs baseline 60.1 (+100%) while TM1_2 stays cold – the remaining 5 motors drawing extra current. n is small for 08-Aug upper buckets; treat as directional, not conclusive.

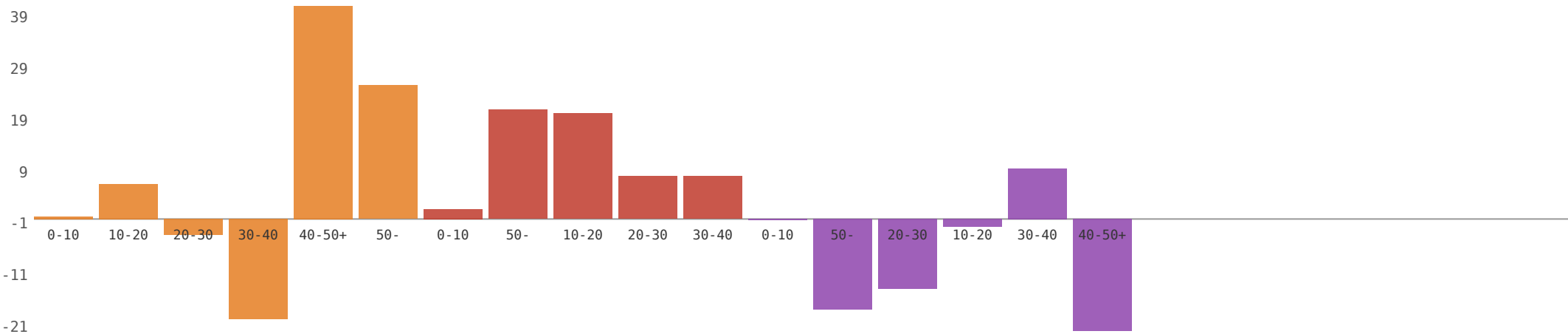
TEST C – axle-04 speed vs the other axles (the kinematic test)



Read (Test C): On 08-Aug pre-failure (top) axle-04 speed tracks axles 1/2/3/5 exactly at every row – the wheelset was **still rotating**, NOT yet seized, even though TM1_2 was cold. On 09-Aug (bottom), the day after the reported 19:26 lock, the axle-04 speed channel reads dead sentinel (3276, dropped from plot) while the other axles show valid crawl speeds. So the axle-04 speed signal failed *after* the incident, not before. No kinematic divergence proves pre-lock binding; the failure moment is inside the 08-Aug 17:25 → 09-Aug 14:25 telemetry gap. (axle-06 channel is dead all days = feed artifact.)

TEST B – primary current residual vs baseline $I=f(v,a)$ model

TEST B - I_excess (A) vs baseline f(v,a) model, per period & speed bin (median)

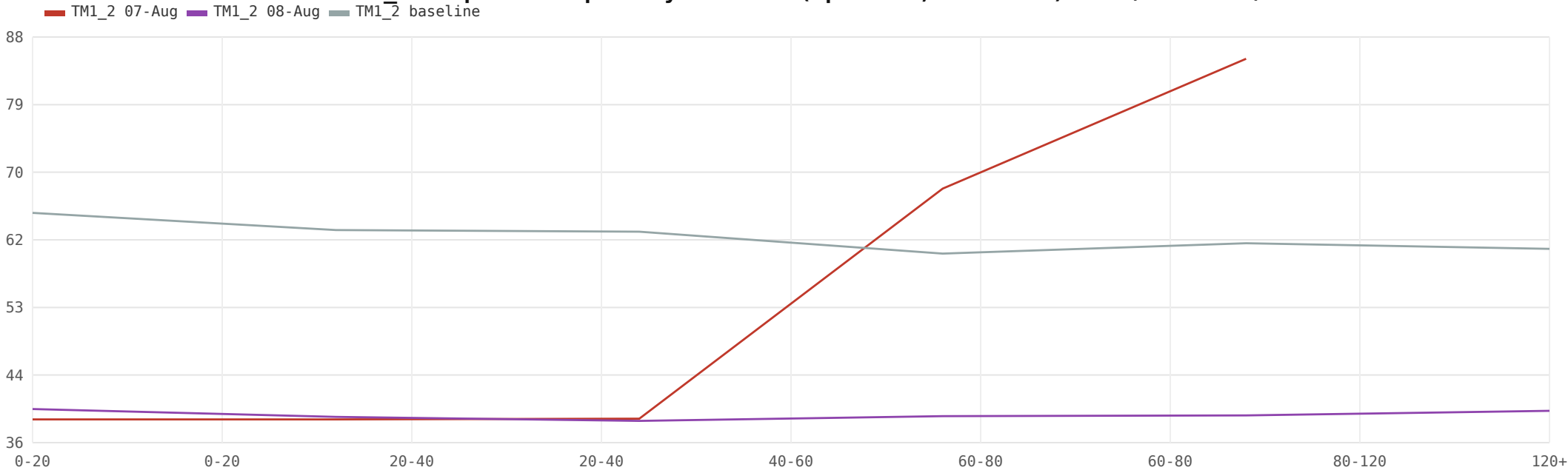


color: 06-Aug=orange, 07-Aug=red, 08-Aug=purple. x = speed bin. bars = median I_excess (A).

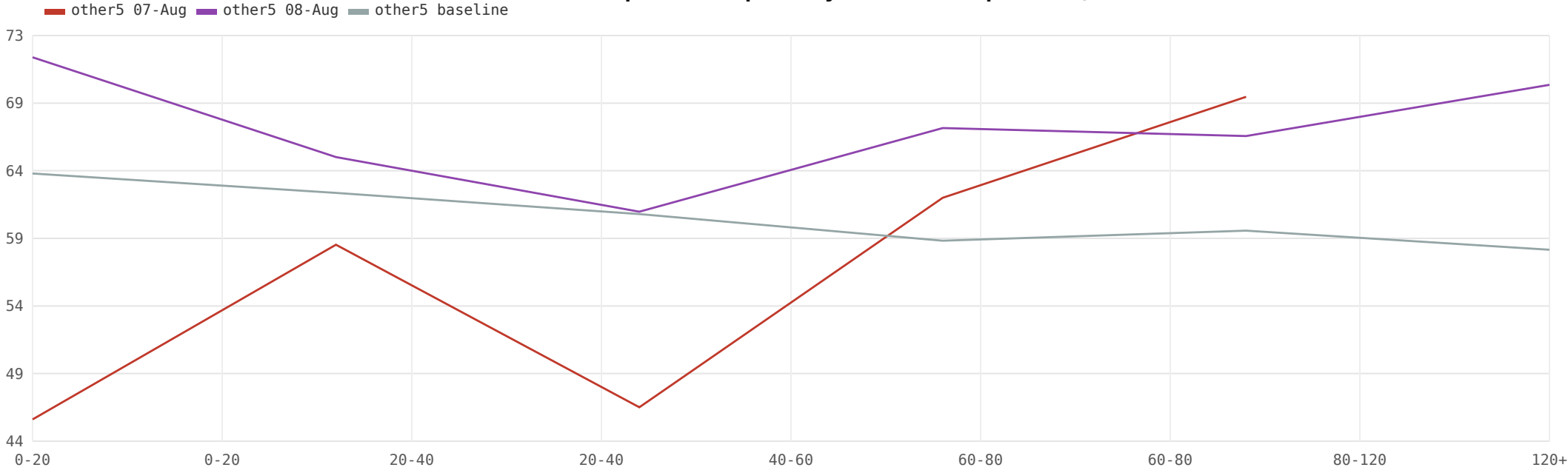
Read (Test B): $I_{\text{excess}} = I_{\text{obs}} - \text{baseline median } I$ at matched (speed, acceleration), built from 27-Jul–03-Aug. 06-Aug (orange) and 07-Aug (red) show **positive excess** in upper speed bins (07-Aug +20.4/+8.3/+21.2 A across bins) – the loco demanded more current than its own baseline at comparable v+a, consistent with increased resistance. 08-Aug (purple) is **not** elevated vs baseline at matched v+a – the train ran no harder; instead the load redistributed (Test E). Small n for 08-Aug upper bins.

TEST E – load redistribution: temp response vs primary current

TEST E - TM1_2 response vs primary current (speed>20, traction) - 07/08 vs 08/08



TEST E - other-5 mean response vs primary current (speed>20, traction)



Read (Test E): On 07-Aug (red) TM1_2 **ris**es with current (60–80 A → 68 °C, 80–120 A → 85 °C) – the anomalous motor still coupled to traction effort. On 08-Aug (purple) TM1_2 is **flat ~38–40 °C at every current level** while the other-5 mean runs 57–72 °C – axle-04 thermally decoupled from current, remaining motors carrying the train. Baseline (grey): all six track together. This is the strongest evidence of the hot-cold state transition.

TEST 6 – energy counter change during traction-active running (direction only)

day	running-min	dCounter	counter/min	vs 05/08
05-Aug	894	2632	2.94	1.0x
06-Aug	877	5492	6.26	2.1x
07-Aug	779	3555	4.56	1.55x
08-Aug	87	1100	12.65	4.3x

Read: the cumulative energy counter (xenergymwh_ec) changes **more rapidly per running-minute** toward the failure (peaking 4.3× the 05-Aug rate on the failure day). Units of the counter are unverified; quote only the relative change, NOT "kWh/min".

TASK 7 - 07/08 19:15 -> 19:50 state transition (event reconstruction)

Ground truth 08/08 19:26 axle-04 lock. This window: 07/08 19:10-19:50. 462 rows (1 s).

Aligned 6-panel plot (markers: green=19:36 first loaded run, red=19:46 dT<-10)



Before / During / After table (medians over each segment)

Signal	Before 19:36	19:36-19:46	After 19:46
TM1_2 (axle04)	39.1	38.9	38.7
other-5 median	38.8	44.9	55.1
dT (TM1_2-other5)	+0.4	-6.1	-16.5
Primary current (A)	9.8	20.4	9.6
Speed (km/h)	0.0	27.6	29.3
axle04 speed (km/h)	0.0	27.0	29.0
Notch xangtrans	0.0	27.8	0.0
LTEDemand	0	1	0
LBEdemand	0	0	0
MVBC on	1	1	1
faultnum	0.0	0.0	0.0

Interpretation

Status / control bits across the whole window (07/08 19:10-19:50): bbur1/2/3_off = 0, bstb1/2_off = 0, bflg1/2_off = 0, bslg1/2_off = 0, bhbb1/2_off = 0, bbdb1/2_off = 0 (all 0 = no isolation/cutout), mtrcctract1/2 = 0, mvcb_on = 1, faultnum = 0 throughout. Only LTEDemand / LBEdemand / xangtrans (notch) change. **KEY EVENT RECONSTRUCTION:**

- 19:15-19:35: idle. v=0, all axle speeds 0, lte=0, xang=0, Ip~8-10 A (aux). TM1_2=39.4, other5=39.0 - all six drifting down together. 19:27:42-19:27:55: brief 15-s shunt 0->2.5 km/h, ALL axles turn together (al=a04=a5=1-2), no divergence. Stops 19:27:59.
- 19:36:42 FIRST LOADED RUN: v 14->33.6 km/h, xang rises to 50 (full notch), Ip up to 81 A. **TM1_2 stays FLAT 38.8-39.0** while other5 rises 42.8->47.1. All axle speeds identical (no axle04 divergence).
- 19:38:12-19:39:27: coast (xang=0, lte=0, Ip 7-10). other5 keeps climbing 44->46 (thermal inertia), TM1_2 flat.
- 19:39:29-19:40:45: second throttle app (xang~26-36, Ip 8->45 A). other5 45.9->47.1, TM1_2 flat 38.7-38.9.
- 19:40:45 -> 19:46:09 **TELEMETRY GAP** (~5.4 min missing). At resume: v=36.1, lte=0, Ip=10.3 (coasting), TM1_2=38.7, **other5=54.6**. The other5 jump 47->55 happened INSIDE the gap (more loaded running).
- 19:46:09-19:46:55: decel, lbe=1 (brake demand) 19:46:32-19:46:52. TM1_2 flat 38.7-38.8, other5 54.6->55.2.
- 19:46:55+: v 28->23 km/h still decelerating; TM1_2 38.7, other5 55.4. bur3 (DC-link current conv3) oscillating 1-73 during decel (regenerative/ dynamic behavior).

INTERPRETATION vs the three hypotheses:

- H-A (sensor fault): NOT supported.** TM1_2 tracked ambient exactly during idle (39.4->38.7 in phase with other5 39.0->38.6), then held a flat ~38.8 through full-notch loaded runs - a live sensor reading a thermally unloaded motor, not a stuck/dead channel (compare axle-06 speed=3276 or xtempmotor3_2 stuck at 76.0 for true dead channels). Also the same channel read 91.7 C that morning.
- H-B (electrical/control isolation): no bit evidence, but consistent.** No isolation/cutout bit toggles in this feed, yet TM1_2's flatness under full-notch load + other-5 carrying the train is exactly what a motor-cut-out / no-contribution state looks like. The feed exposes no per-motor cutout bit, so absence of telemetry is NOT proof either way.
- H-C (mechanical): the morning overhear (91.7 C) + current-positive residual fits rising friction; the flat-TM + normal-aggregate-current + still-rotating-axle fits a motor that has stopped contributing load.**

CONCLUSION (what changed at 19:36 / 19:46): TM1_2 was ALREADY thermally decoupled at the FIRST loaded run of this window (19:36) - it never heated under load while the other five did. The "19:46 transition" is a dT<-10 threshold crossing driven by the OTHER motors' temperature reaching ~55 C (partly inside the 19:40-19:46 data gap), NOT a separate axle-04 event. No status/control/fault bit changed in the window; axle-04 kept rotating at train speed throughout. The cleanest

statement: after the 07/08 morning overheat, axle-04's motor stopped responding thermally to traction load - a loss-of-thermal-contribution state, whose mechanism (electrical cutout vs mechanical) cannot be separated with the signals captured here.